Computer Graphics

Viewport Transformation

3D Viewing

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1

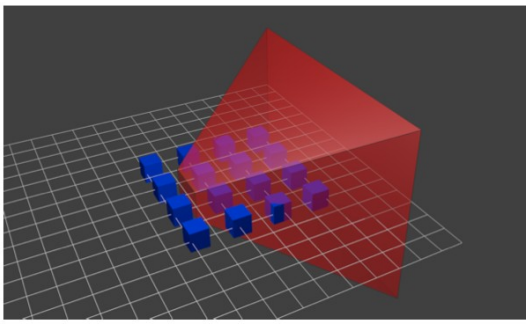
Canonical View Volume

- Projection is not performed right away; instead, map the view volume to a cube of fixed dimensions, called the canonical view volume or a standard view volume
- A normalizing matrix performs this transformation.
- Why?
  - Easier to eliminate objects outside the view volume.
  - Orthographic & perspective aren't different.
  - The z-coordinates not thrown away. (Used later!)

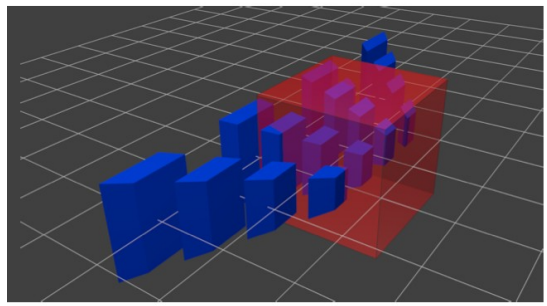
2

Canonical View Volume Visualisation

Perspective View Volume



Canonical View Volume



http://www.opengl-tutorial.org/beginners-tutorials/tutorial-3-matrices/

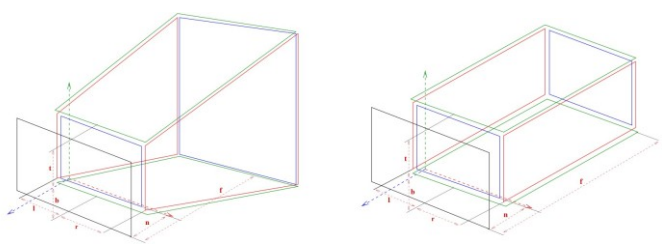
3

Canonical View Volume Dimensions

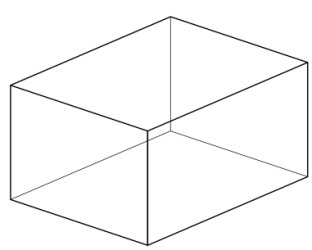
- OpenGL:

$$-1 \leq x \leq 1$$
$$-1 \leq y \leq 1$$
$$-1 \leq z \leq 1$$

- Canonical view volume is the Orthographic View Volume, with appropriate scaling.



To target view volume:



4

Orthographic Normalising Matrix

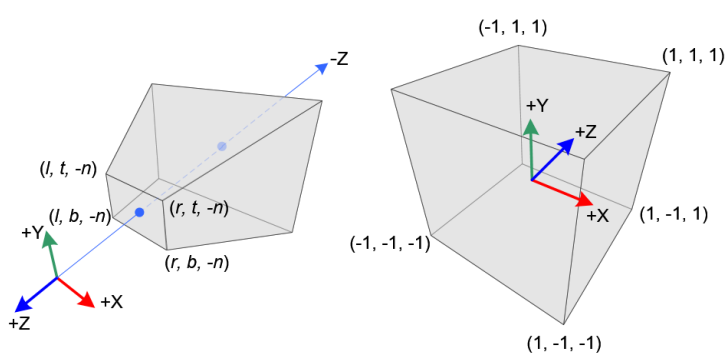
- Lengths (right - left), (top - bottom) and (far - near) scaled to 2
- Shift origin

$$\begin{bmatrix} \frac{2}{right - left} & 0 & 0 & -\frac{right + left}{right - left} \\ 0 & \frac{2}{top - bottom} & 0 & -\frac{top + bottom}{top - bottom} \\ 0 & 0 & \frac{2}{far - near} & \frac{far + near}{far - near} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Drop *z* and use the (x, y) coordinates as the (normalized) window coordinates.

Perspective Projection

- Given:
  - Near Plane
  - Far Plane
  - X and Y bounds
- Compute:
  - Projection transform considering the view volume



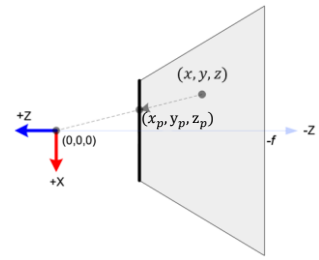
https://www.songho.ca/opengl/gl_projectionmatrix.html

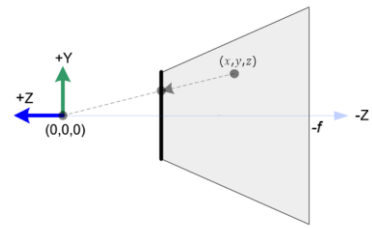
Considering x and y only

Perspective Projection

- Classical perspective projection projects (x, y) coordinates to (x_p, y_p) and drops z coordinates
- $\frac{x_p}{-\mathrm{N}} = \frac{x}{z}, x_p = -\frac{\mathrm{N}}{\mathrm{Z}} x$
- $\frac{\mathrm{y}_p}{-\mathrm{N}} = \frac{\mathrm{y}}{z}, \mathrm{y}_p = -\frac{\mathrm{N}}{\mathrm{Z}}\, y$
- Note that x_p and y_p are inversely proportional to $-z$

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$



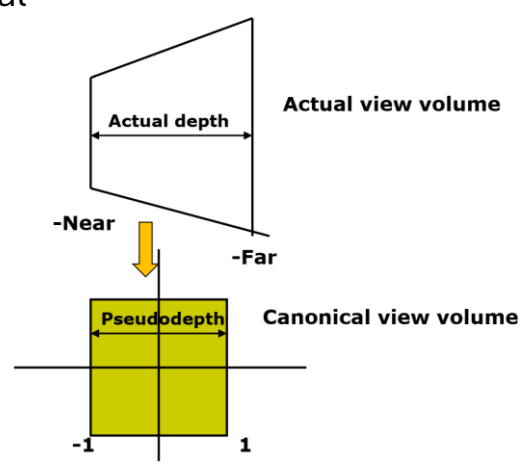


Considering z

Perspective Projection

- **Perspective transformation** maps actual z distance of perspective view volume to range [–1 to 1] (**Pseudodepth**) for canonical view volume
- Set scaling to $-z$
- Pseudodepth $= az + b$
- $\mathrm{z}_p = \frac{\mathrm{az+b}}{-\mathrm{z}}$

$$M_1 = \begin{bmatrix} N & 0 & 0 & 0 \\ 0 & N & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & -1 & 0 \end{bmatrix}$$



Solving for a and b

Perspective Projection

- Given $z_p = \frac{az+b}{-z}$, use boundary conditions
  - $z_p = -1$ when $z = -N$
  - $z_p = 1$ when $z = -F$
- $a = \frac{-(F+N)}{F-N}$, $b = \frac{-2FN}{F-N}$

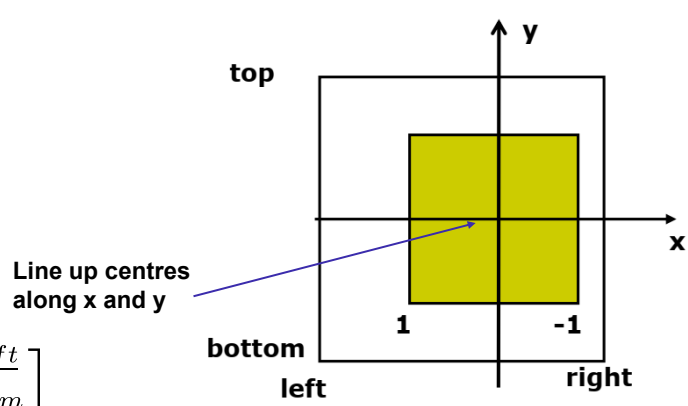
$$M_1 = \begin{bmatrix} N & 0 & 0 & 0 \\ 0 & N & 0 & 0 \\ 0 & 0 & -\frac{F+N}{F-N} & -\frac{2FN}{F-N} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

- Can transform z, but x and y needs to be bounded between [-1,1]

Translate in x and y

Perspective Projection

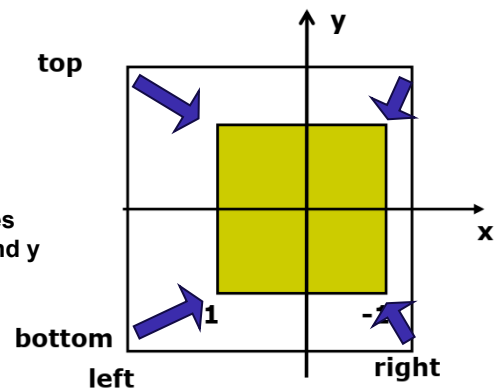
- We translate by
  - X: $-(right + left)/2$
  - Y: $-(top + bottom)/2$



$$M_2 = \begin{bmatrix} 1 & 0 & 0 & -\frac{right+left}{2} \\ 0 & 1 & 0 & -\frac{top+bottom}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scale in x and y

- Scale by:
  - $2/(right - left)$ in x
  - $2/(top - bottom)$ in y



$$M_3 = \begin{bmatrix} \frac{2}{right-left} & 0 & 0 & 0 \\ 0 & \frac{2}{top-bottom} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final Perspective Projection Matrix

$$\begin{bmatrix} \frac{2}{right-left} & 0 & 0 & 0 \\ 0 & \frac{2}{top-bottom} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -\frac{right+left}{2} \\ 0 & 1 & 0 & -\frac{top+bottom}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} N & 0 & 0 & 0 \\ 0 & N & 0 & 0 \\ 0 & 0 & -\frac{F+N}{F-N} & -\frac{2FN}{F-N} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

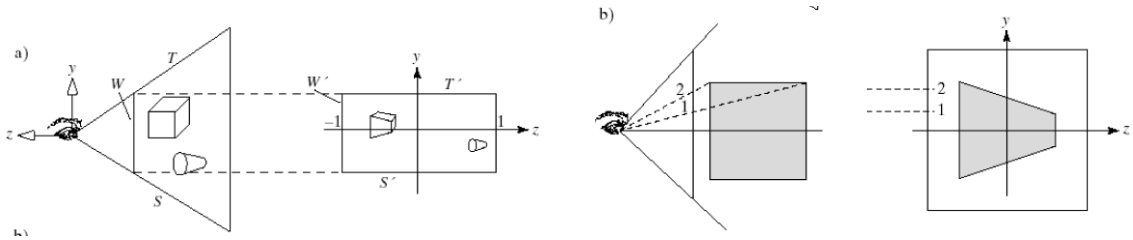
$$\begin{bmatrix} \frac{2N}{right-left} & 0 & \frac{right+left}{right-left} & 0 \\ 0 & \frac{2N}{top-bottom} & \frac{top+bottom}{top-bottom} & 0 \\ 0 & 0 & -\frac{F+N}{F-N} & -\frac{2FN}{F-N} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

glFrustum(left, right, bottom, top, N, F)

Geometric Nature of Perspective Transform

Lines through eye map into lines parallel to z axis after transform

Lines perpendicular to z axis map to lines perpendicular to z axis after transform



Final 2D Coordinates in NC

$$(u, v, d) \equiv \begin{bmatrix} x_p \\ y_p \\ z_p \\ w \end{bmatrix} = M_3 M_2 M_1 \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Perspective division: Divide x_p, y_p coordinates by the w to get the normalized coordinates (u, v)
- z_p maintains ordering and can be used without division
- The normalized d component has non-linear precision. Higher around the near plane and lower around the far plane due to the division by z.

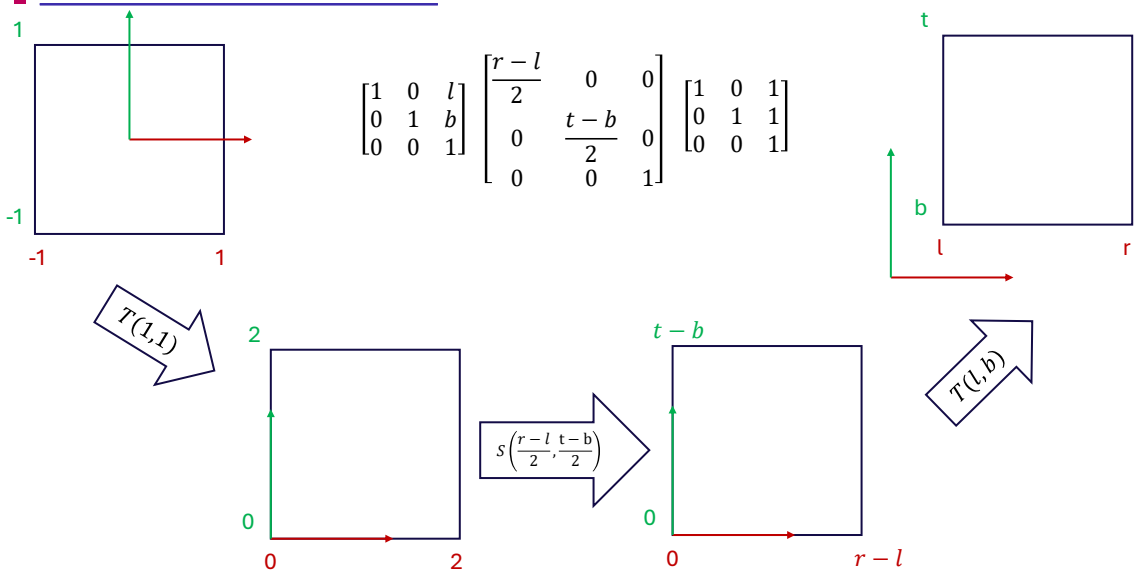
Actual Projection

- We have already performed the perspective division.
- Projection involves simply dropping the z coordinate and scaling x-y to the viewport.
- Why go through with the z-coordinates?
- The ordering is preserved along the depth dimension.
- z values can be used for visibility determination.

Where is the Film ?

- Turns out: It does not matter.
- A final scaling is in the viewport transformations.
- As long as the film is in front of the camera, we will see an upright image.
- Can consider the near plane as the film plane.

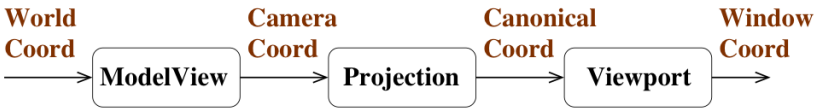
Viewport Transform



$$\begin{bmatrix} 1 & 0 & l \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{r-l}{2} & 0 & 0 \\ 0 & \frac{t-b}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Graphics Pipeline in Action

- Points are transformed from Object to World to Canonical to Window coordinates.
- Each 3D point maps to a pixel (i, j) in the window space.
- Lines are made out of two points. Triangles and polygons are made out of 3 or more points.



Summary

- Camera can be perspective or parallel (orthographic, oblique). 6 planes give the view volume and defines the camera.
- View Mapping involves mapping the world to a canonical view volume, which is an orthographic view volume. This Normalizing Transformation has different forms for parallel and perspective cameras.
- Projection and Clipping are easy to perform in the canonical view volume. An image with dimensions from -1 to +1 results.
- Viewport transformation is the final step, involving a 2D scaling and translation to map to window coordinates that can be used to address the frame buffer.

Thank You

Symmetric Perspective Transform

Field of View